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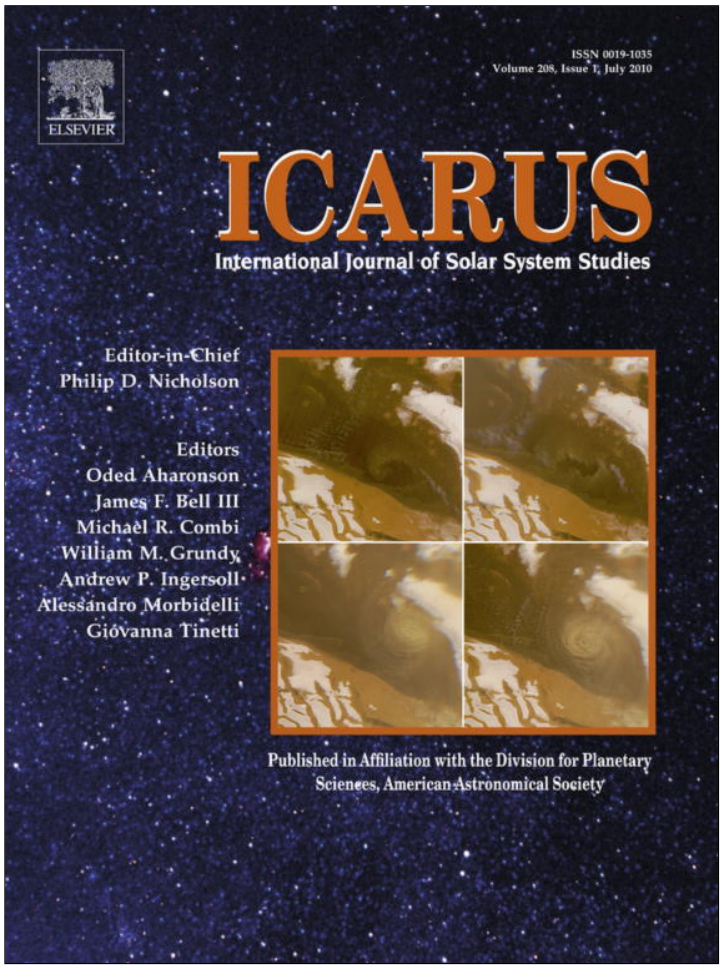

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The thermal conductivity of meteorites: New measurements and analysis

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ABSTRACT

We have measured the thermal conductivity at low temperatures (5–300 K) of six meteorites representing a range of compositions, including the ordinary chondrites Cronstad (H5) and Lumpkin (L6), the enstatite chondrite Abee (E4), the carbonaceous chondrites NWA 5515 (CK4 find) and Cold Bokkeveld (CM2), and the iron meteorite Campo del Cielo (IAB find). All measurements were made using a Quantum Design Physical Properties Measurement System, Thermal Transport Option (TTO) on samples cut into regular parallelepipeds of ~2–6 mm dimension. The iron meteorite conductivity increases roughly linearly from 15 W m^{-1} K^{-1} at 100 K to 27 W m^{-1} K^{-1} at 300 K, comparable to typical values for metallic iron. By contrast, the conductivities of all the stony samples except Abee appear to be controlled by the inhomogeneous nature of the meteorite fabric, resulting in values that are much lower than those of pure minerals and which vary only slightly with temperature above 100 K. The L and CK sample conductivities above 100 K are both about 1.5 W m^{-1} K^{-1}, that of the H is 1.9 W m^{-1} K^{-1}, and that of the CM sample is 0.5 W m^{-1} K^{-1}; by contrast the literature value at 300 K for serpentine is 2.5 W m^{-1} K^{-1} and those of enstatite and olivine range from 4.5 to 5 W m^{-1} K^{-1} (which is comparable to the Abee value). These measurements are among the first direct measurements of thermal conductivity for meteorites. The results compare well with previous estimates for meteorites, where conductivity was derived from diffusivity measurements and modeled heat capacities; our new values are of a higher precision and cover a wider range of temperatures and meteorite types. If the rocky material that makes up asteroids and provides the dust to comets, Kuiper Belt objects, and icy satellites has the same low thermal conductivities as the ordinary and carbonaceous chondrites measured here, this would significantly change models of their thermal evolution. These values would also lower their thermal inertia, thus affecting the Yarkovsky and YORP evolution of orbits and spin for solid objects; however, in this case the effect would not be as great, as thermal inertia only varies as the square root of the conductivity and, for most asteroids, is controlled by the dusty nature of asteroidal surfaces rather than the conductivity of the material itself.

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1. Introduction

The thermal properties of stony meteorites are an important fundamental physical characteristic of these materials, an indication of both their chemical and physical natures. Furthermore, knowing these thermal values can put important constraints on the thermal response of asteroids to heating from the Sun, an important parameter in the Yarkovsky and YORP effects on asteroid orbital and spin perturbations. But perhaps most importantly, the thermal evolution both of asteroids and of comets, icy satellites, and other icy bodies thought to have a significant meteorite-like rocky component will obviously depend on the thermal properties of their constituent materials, for which meteorites are our best known analogs.

The Yarkovsky effect describes how an asteroid's orbit can gain or lose energy due to the infrared re-radiation of absorbed sunlight; the Yarkovsky–O'Keefe–Radzievskii–Paddack (YORP) effect describes how this re-radiation alters the spin properties of the asteroid. Both depend on the *thermal inertia* of the body, which can cause a stronger flux of infrared energy to come from the afternoon side of the spinning asteroid. The thermal inertia is defined as $\sqrt{(\rho C k)}$, where ρ is the density of the material, C is the heat capacity, and k is the thermal conductivity. (In calculations, one commonly uses the *inverse thermal inertia*, often designated by the Greek letter Γ.)

The thermal evolution of a body, on the other hand, can be thought of as a diffusion process, with a *thermal diffusivity* κ defined as $(k\rho^{-1}C^{-1})$, the ratio of the thermal conductivity to the volumetric heat capacity. This term is linearly related to the thermal conductivity, which is the fundamental property of a material's ability to conduct heat. Thus, more so than in the case of thermal

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inertia, a factor of two or greater change in thermal conductivity can have a profound effect on the expected thermal evolution of a small body.

Note that the same three quantities define both thermal inertia and thermal diffusivity: density (ρ), heat capacity (C), and thermal conductivity (k). In recent years, the densities (mass per unit volume) of more than a thousand different meteorites have been measured (for a recent review see Consolmagno et al., 2008), and a good understanding exists of the typical ranges of densities for most meteorite types. Within a particular class of meteorite, the variation in density is less than 10% and the spread in porosity is generally confined to a range of about 10% above or below the average value for a given class. Independent of class, most stony meteorites (except hydrated carbonaceous chondrites) can be presumed to have a density of about 3.5 g cm^{-3} and a porosity of about 10%, though several carbonaceous chondrite classes have significantly higher porosities.

The heat capacity (C) is the ability of a given mass of a substance to store internal energy while undergoing a specified temperature change without undergoing a phase change; it is the measure of the heat energy required to change the temperature of a unit mass of a material. While heat capacity has not been measured for a wide range of meteorites, a few values do exist in the literature and they track reasonably well with the heat capacity predicted from literature values for the constituent materials of these meteorites. This is not surprising, since heat in a crystal is stored in the individual molecular bonds and these bonds are not strongly affected by the size or arrangement of the individual minerals within a meteorite. Furthermore, the variation of heat capacities among common minerals is rarely more than 20%. A typical value chosen for many thermal models is around 750 J kg^{-1} K^{-1} (cf. the discussion in Ghosh and McSween (1999)), in agreement with at least one set of ordinary chondrite measurements at room temperature (Beech et al., 2009). Yomogida and Matsui (1983) measured the thermal diffusivity of 20 ordinary chondrite meteorites and used these data to calculate heat capacities, coming to similar values at 300 K but noting that, assuming a variation with temperature consistent with that measured for the minerals, one should expect meteorite heat capacities to drop to about 500 J kg^{-1} K^{-1} at 200 K. However, earlier work by this group (Matsui and Osako, 1979) had directly measured the heat capacity of five Yamato meteorites (four ordinary chondrites and a howardite) at 300 K, 350 K, and 400 K with results only about two-thirds of these theoretical values. Interestingly, in their later paper they appear to prefer the theoretical values. Clearly more measurements would be useful here, and a campaign of measuring the heat capacities of meteorites over a range of temperatures is planned for future work.

This paper will concentrate on the measurement of the third factor in these quantities, the thermal conductivity.

2. Previous estimations of meteorite thermal conductivity

Thermal conductivity is the most uncertain constituent of the diffusivity and inertia values, and the most problematic to measure. For meteorites, thermal conductivity is largely unknown; direct measurements of thermal conductivities for stony meteorites are difficult to find in the literature. Essentially the only systematic work to date is that of Matsui and Osako (1979) and Yomogida and Matsui (1983), who measured the thermal diffusivity of a number of meteorites and then from these values calculated thermal conductivities for a total of 22 different meteorites (some samples are duplicated between the two papers). Their method was to find the product of their measured diffusivity and density values with, for most samples, a heat capacity calculated from the meteorite's mineralogy. There is a significant spread in their results. For example, their thermal conductivity values for different ordinary chondrites at 200 K range over nearly an order of magnitude, from 0.4 to 3.8 W m^{-1} K^{-1}, with L chondrite values only slightly lower, and mostly overlapping, those of H chondrites. In addition, a recent paper by Beech et al. (2009) has measured the thermal conductivity of the Gao Guernie H5 chondrite, reporting a value of 3.0 W m^{-1} K^{-1} at room temperature. (The value reported in that paper is 10 times too big, due to a units conversion mistake, Beech, personal communication.) All these values are significantly lower than the well-measured conductivities of olivine, pyroxene, or plagioclase.

Perhaps because of such problems, most thermal modelers have not taken advantage of these measurements but have merely assumed that meteorite thermal conductivity is similar to that of the more common meteoritic minerals, such as olivine, pyroxene, or serpentine. In some cases a correction is made to account for the porosity within the material; more rarely is the variation of these properties with temperature also taken into account. However, as Ghosh and McSween (1999) pointed out in the case of heat capacity, the variation of all these thermal quantities at temperatures relevant to the outer Solar System can have important effects on thermal models.

Solid state theory states that heat is transported across insulating crystals by the passage of packets of mechanical vibrations akin to sound waves called *phonons*. There are several limits on the passage of phonons and thus the thermal conductivity of a solid material. At very low temperatures (generally below 100 K, down to temperatures approaching absolute zero) a number of vibrational modes within the crystal are suppressed; theory thus predicts that within this range the resultant conductivity will vary as the cube of the temperature. At higher temperatures phonons will begin to interfere with each other, resulting in a suppression of conductivity and this effect increases as the temperature increases. Thus, above about 200 K, one predicts that the thermal conductivity will fall as $1/T$. Combining these effects, one expects a thermal conductivity in a typical crystal should rise sharply at temperature increases from absolute zero to some maximum value between 100 K and 300 K, followed by a $1/T$ decrease as temperature increases. This pattern is, in fact, seen in our measurements of the E chondrite Abee.

Besides scattering off each other, phonons can also be strongly scattered by inhomogeneities within the fabric of the material. Such obstacles to phonon passage include pore spaces, density contrasts, and grain boundaries. For the chondrites, the presence of many small grains of metal should also contribute to the scattering of phonons and thus the suppression of thermal conductivity. With these many competing effects, predicting the thermal conductivity of asteroidal material from first principles is uncertain at best. For this reason, we have undertaken to collect reliable laboratory measurements using state of the art equipment of the thermal conductivity of meteoritic material at temperatures of interest for the asteroid belt and the outer Solar System.

3. Technique

Measuring thermal conductivity is difficult. As noted in a recent paper by Hofmeister and Pertermann (2008), "contact" methods that impart a pulse of heat into the sample depend on making a good thermal contact between the sample and the source of the heating (and the sensor measuring the response of the sample to the heating), which is difficult; a poor contact will result in underestimating the k value of the material being measured. On the other hand, a competing method which provides heat via a pulse of a laser onto the surface of the material is open to serious error if the initial transfer of energy through the crystal by laser photons,

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rather than phonons, is not adequately accounted for. This has the opposite effect of overestimating k.

For our measurements, we used a Quantum Design Physical Property Measurement System, Thermal Transport Option (PPMS–TTO), a modern system designed to overcome the inherent difficulties noted above. The TTO system allows two measurement modes, continuous measurement mode and single-step mode. Our results were obtained by measurements in continuous mode. Thermal conductivity is determined by applying a heat pulse "Q" by a heater attached to one end of the sample in order to create a user-specified temperature difference between two calibrated Cernox thermometers located at the sample ends. Heat flows out of the sample into a cold-foot located on the sample puck. The samples were attached via silver (Ag) epoxy to gold-coated oxygen free high conductivity copper (OFHC-Cu) disks which allow for good thermal contact, good heat flow, and a highly reproducible method for attaching samples and thermometers. A gold-coated copper shield plate isolates the sample from the other parts of the TTO assembly and minimizes radiation effects. A cylindrical, copper brass-coated copper shield screws into the base of the puck and is designed to minimize thermal radiation from the sample environment. A mounted meteorite sample (Cronstad) is shown in Fig. 1. The sample puck is held in a cryostatic chamber whose temperature is automatically stepped from 300 K to below 5 K at a rate of 0.5 K/min, in a vacuum (pressure is held to $<1.33 \times 10^{-4}$ Pa). The accuracy of the measurements with this Ag epoxy has been confirmed on a 7740 Pyrex standard, and temperature calibrations were performed using the Quantum Design Ni-alloy standard (Dilley et al., 2002).

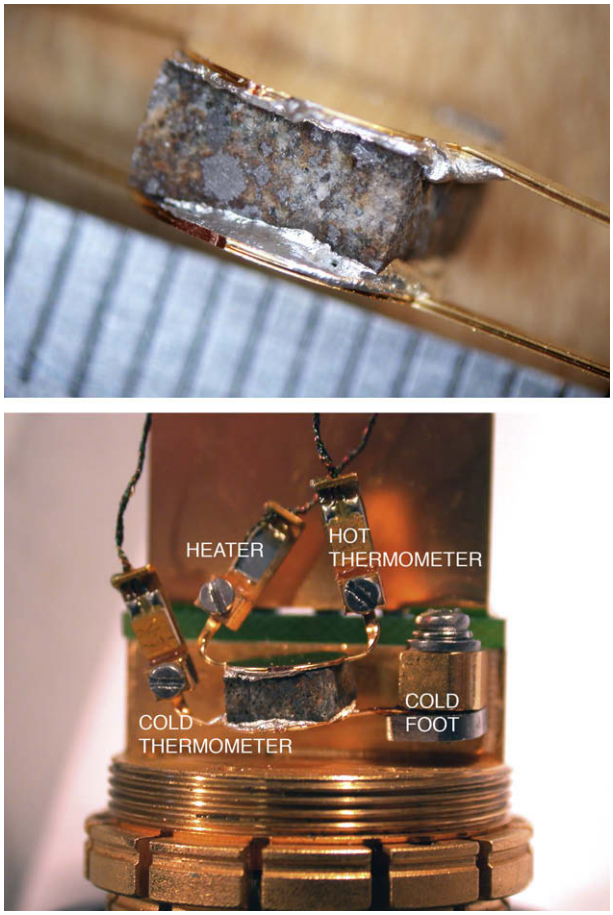

Fig. 1. Cronstad (H chondrite) sample, mounted in the sample chamber of the Quantum Design P670 TTO used in the measurements of thermal conductivity.

Six different meteorites, spanning the most common types of meteoritic material, were chosen for these measurements.

For the ordinary chondrites, the H chondrite Cronstad and the L chondrite Lumpkin were measured. Cronstad is a veined H5 which fell in South Africa in 1877; its olivine iron content is Fa_{18}, total iron 26.65%. Lumpkin is an L6 that fell in the southeastern US in 1869. Both samples show evidence of minor weathering. (Originally classified as an L, Lumpkin's olivine Fe continent – Fa_{19}, similar to an H chondrite – led Mason (1963) to reclassify it as an H, and this was followed in several subsequent catalogs of meteorites. However, Chou et al. (1973) showed that both its Fe metal content – less than 10% – and Ni, Ga, Ge, and Ir contents all support its original classification as an L; likewise, the Meteoritical Bulletin website prefers the L classification. As we show here, its thermal properties are significantly different from the H chondrite Cronstad, in the direction expected for a sample with lower metal content.) The enstatite chondrite Abee (30.35% total iron, less than 1% Fs) fell in Canada in 1952. Two carbonaceous chondrites were measured: NWA 5515, collected in the Algerian desert in 2007, a CK4 (Fa_{29-34}); and Cold Bokkeveld, a CM2 that fell in South Africa in 1838, with 20% total iron content, and containing carbonates and aqueous alteration. Finally, a piece of the IAB iron meteorite Campo del Cielo (found in Argentina, 1576) was also measured. All but the Campo del Cielo piece were provided from the Vatican Observatory collection. (Data from Grady (2000) and the *Meteoritical Bulletin* on-line databank.)

Each sample was cut into a regular parallelepiped whose dimensions lay between 2 and 6 mm (see Table 1). For most samples, the cutting and mounting process mean that there would be times when the samples were placed in acetone (to dissolve excess epoxy) or cured in an oven at a temperature of 373 K. However, since such a procedure could alter the structure of the hydrated CM meteorite, special care was taken with this sample; rather than being mounted to a plate for cutting it was held by hand while being cut, and the thermal epoxy was cured at room temperature over 24 h rather than in an oven.

Given the regular shapes of the cut samples it is possible to calculate directly the bulk densities of each sample, as is shown in Table 1. It is interesting to note the differences between the densities and porosities calculated here and those typically measured for larger bulk samples of a given class. In every case, as shown in Table 1, the inferred porosity for our small samples is larger than the porosity typical for meteorites of that class. The largest difference is seen in Lumpkin. It is possible that the cutting process itself has introduced porosity into the samples. (Notice the voids visible in the side of the sample in Fig. 1, of a size and shape not normally seen in meteorite thin sections.) Since, as we argue below, porosity is what controls the thermal conductivity, this effect could be an important source of error. On the other hand, these voids may be confined to the edges of the sample and thus have little effect on the thermal properties of the sample as a whole. In any case, we note that these higher inferred porosities still lie within the range of porosities actually observed in meteorite samples.

4. Results

The thermal conductivities of the six meteorites measured are illustrated in Fig. 2. Their values at 200 K and the derived value for the thermal diffusivity κ and inverse thermal inertia Γ at that temperature for an assumed value of the heat capacity C are presented in Table 2.

The iron meteorite Campo del Cielo not surprisingly shows a very different thermal conductivity from the stony meteorites. We note the slight deviation of the measured thermal conductivity at 150–170 K; this may be due to the thermal contraction and sub-

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Table 1
Sample characteristics.

Meteorite	Size (mm)	Mass (g)	Density (g cm^{-3})	Grain, ρ^a (g cm^{-3})	Model porosity (%)	Class ave. porosity (%)
Abee (E4)	3.653 × 4.853 × 3.583	0.2083	3.279	3.6	8.9	5
Campo del Cielo (IAB)	3.590 × 3.063 × 2.793	0.2368	7.710	7.8	1.2	0
Cold Bokkeveld (CM2)	3.006 × 2.062 × 5.998	0.0618	1.662	2.65	37.3	25
Cronstad (H5)	5.224 × 2.759 × 5.753	0.2612	3.150	3.78	16.7	10
Lumpkin (L6)	3.9624 × 5.080 × 3.7084	0.2185	2.927	3.62	19.1	7
NWA 5515 (CK4)	2.574 × 3.024 × 3.199	0.0666	2.675	3.57	25.1	23

[a] Taken from averages of unweathered meteorites of given class. Cold Bokkeveld grain density from Mason, personal communication.

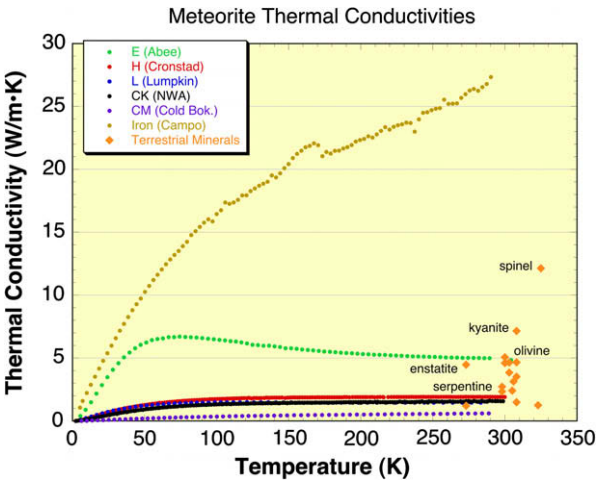

Fig. 2. Summary of meteorite thermal conductivity measurements, with assorted conductivities for terrestrial minerals taken from Clauser and Huenges (1995).

sequent removal of cracks within the sample, resulting in a slight increase in the thermal conductivity at that point. Above 100 K, the conductivity can be fit with a line of the form $k = 12.4 + 0.05\ T$, with an R^2 value of 0.94.

The thermal conductivity of the enstatite chondrite Abee shows the expected T^3 increase at low temperatures and $1/T$ decrease at high temperatures. Above 100 K, the conductivity is fit very well with the formula $k = 4.11 + 248/T$, with an R^2 value of 0.998. At 300 K the conductivity is close to that of olivine and enstatite; apparently the increased conductivity expected from the iron content is effectively balanced by the decreased conductivity caused by the sample's porosity and mineral inhomogeneity. It is important to note that Abee is a breccia made of both metal-rich and metal poor clasts (Sears et al., 1983), and these clasts can be much larger than the sample measured here. Indeed, enstatite meteorites in general are known to be heterogeneous on scales larger than 10 g (Jarosewich, 1990). Thus our result may not be representative of the whole meteorite, or of the enstatite meteorite parent body in general.

The conductivities of the other stony meteorites are shown in more detail in Fig. 3. From 100 K to 300 K, the thermal conductivities of the ordinary chondrites Cronstad and Lumpkin are essentially constant; Cronstad only increases from 1.75 to 1.93 W m^{-1} K^{-1} (with most of that change occurring between 100 K and 150 K) while Lumpkin ranges only between 1.46 and 1.51 W m^{-1} K^{-1} (the peak occurring at 120 K, the conductivity decreasing slightly from there to 300 K). The average value for Cronstad over this range is 1.88 W m^{-1} K^{-1} and that of Lumpkin is 1.47 W m^{-1} K^{-1}; both averages are close to the 200 K values reported in Table 2.

The carbonaceous samples vary somewhat more over this temperature range; from 100 K to 300 K their thermal conductivities increase linearly with temperature. The dry CK meteorite NWA 5515 (mineralogically similar to meteorites of the CV class) has a conductivity that can be fit by the equation $k = 1.26 + 0.0011\ T$, with an R^2 value of 0.9; its average value over this range is 1.48 W m^{-1} K^{-1}, similar to its value at 200 K. The hydrated CM meteorite Cold Bokkeveld conductivity from 100 to 300 K is fit by the formula $k = 0.26 + 0.0013\ T$, with an R^2 value of 0.99; its average value in this range is 0.50, again matching its value at 200 K. As it happens, all the data for Cold Bokkeveld down to 6.25 K can be very well fit by $k = -0.0254 + 0.00563\ T - 2.07 \times 10^{-5}\ T^2 + 3.11 \times 10^{-8}\ T^3$, with an R^2 value of 0.9998. This empirical formula may be of use in thermal models of comet nuclei, Kuiper Belt objects, and other outer Solar System objects.

The results for the stony meteorites are shown in greater detail in Fig. 3. This figure also shows, in shaded red and blue areas (with the overlap in purple), the range of thermal conductivities derived for H and L chondrites by Yomogida and Matsui (1983). Note that our measurements generally fit well within this range. We confirm their observation that ordinary chondrites are significantly less conductive than one would calculate simply from the constituent minerals. More data on different ordinary chondrites will allow us to test whether their large range of conductivities, spanning an order of magnitude, is real or an artifact of their measurement technique.

5. Discussion

In our initial discussion on the transport of heat in an insulating material, we noted the role that increased porosity might have in attenuating the passage of phonons in meteoritic material, resulting in lower thermal conductivity. Indeed, Yomogida and Matsui

Table 2
Measured and derived properties at 200 K.

Meteorite	Density (g cm^{-3})	k, 200 K (W m^{-1} K^{-1})	C,[a] 200 K (J kg^{-1} K^{-1})	$\kappa \times 10^7$ (m^2 s^{-1})	$\Gamma \times 10^4$ (m^2 s$^{1/2}$ kJ^{-1})
Abee (E4)	3.279	5.35	500	32.63	3.38
Campo del Cielo (IAB)	7.71	22.4	375	77.48	1.24
Cold Bokkeveld (CM2)	1.662	0.5	500	6.02	15.51
Cronstad (H5)	3.15	1.88	550	10.85	5.54
Lumpkin (L6)	2.927	1.47	570	8.81	6.38
NWA 5515 (CK4)	2.675	1.48	500	11.07	7.11

[a] Adapted from data and calculations in Yomogida and Matsui (1983).

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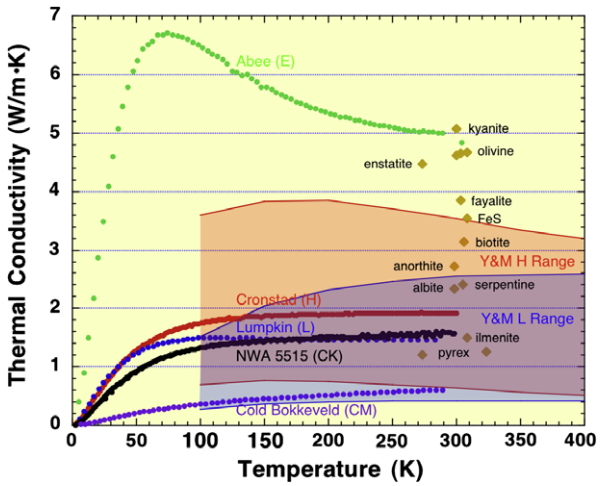

Fig. 3. An expanded version of Fig. 2, showing the thermal conductivity of ordinary and carbonaceous chondrites in greater detail. The red and blue shaded areas (including the purple overlap) show the range of conductivities for H and L chondrites, respectively, reported by Yomogida and Matsui (1983). Terrestrial mineral conductivities are from Clauser and Huenges (1995).

(1983) reported a rough correlation between conductivity and porosity in their data. Certainly the low conductivities reported here are consistent with such an effect. Flynn et al. (1999) and Flynn (2004) showed that the speed of sound in meteorites is also much lower in porous meteorites than is typical for solid rock samples. In addition, those authors noted that certain extremely porous meteorites (for example, the L4 chondrite Saratov) are riddled with large cracks that may not be present in the small samples measured here. Such extremely friable ordinary chondrites are rare in our collections, but not necessarily rare in space; our measurements may well be biased in favor of the stronger, and thus more conductive, samples that survive passage through the Earth's atmosphere. We should expect rocky material in small Solar System bodies to have a low thermal conductivity.

The greatest impact these values for meteorite thermal conductivities will be in the modeling of small body thermal evolution. For a long time, typical thermal history models for asteroids or icy moons (cf. Consolmagno, 1975; Cohen and Coker, 2000) have assumed that the conductivity of the rocky component is similar to that of serpentine, a mineral known to exist in hydrated meteorites and whose thermal conductivity is already significantly lower than other rocky materials. But, as we have seen, the actual conductivity of our hydrated CM meteorite is four times lower than that of laboratory-grade serpentine. And, indeed if conductivity decreases with increased porosity, then one might expect that even more highly porous meteorites such as Orgueil and Tagish Lake would have even lower thermal conductivities.

The thermal diffusivity of stony meteorites is significantly different from that assumed by recent thermal models of asteroids or the rocky component of icy bodies. Cohen and Coker (2000), modeling the evolution of hydrous asteroids, assumed that thermal conductivity would be independent of temperature and they began values of 5.155 and 2.95 W m^{-1} K^{-1}, appropriate for forsterite and serpentine respectively. When they attempted to correct for the effect of porosity, they came up with a value of 2.8 W m^{-1} K^{-1} for the carbonaceous chondrite component in their models. This is still nearly twice our measured value for our CK meteorite and more than five times greater than our measurement for our CM meteorite.

Similarly, these values can affect models for bodies thought to be a mixture of rock and ice, such as the satellites of the outer Solar System planets or the trans-neptunian objects. Schubert et al. (2007) assume a rock conductivity of 3 W m^{-1} K^{-1} in their models of Enceladus. Prialnik and Merk (2008) assume a thermal conductivity of 2 W m^{-1} K^{-1} for the dusty component in their models of Kuiper Belt objects and Enceladus. Even at 100 K – much warmer than their ambient temperatures, but perhaps applicable to warmer interiors – this value would be about 33% too high if the dust is assumed to be similar to anhydrous carbonaceous chondrite materials, and a factor of four too high if it is similar to CM material. For the ambient temperatures found in the Kuiper Belt, this value is a full order of magnitude too high compared to the measured thermal conductivity of the CM meteorite.

A recent thermal model of Kuiper Belt objects by Desch et al. (2009) has noted this problem, and they reference the earlier Yomogida and Matsui (1983) work in choosing a lower value of 1–2 W m^{-1} K^{-1} for the thermal conductivity of the rocky component in their models. However, even here they assume that the conductivity is constant with temperature. As we have shown, even this low conductivity is higher than that of the hydrated meteorites at all temperatures of interest, and much too high for all materials at the ambient temperatures of the Kuiper Belt, below 100 K.

Recall that the thermal diffusivity is linearly related to the thermal conductivity, and its units are meters squared per second. Thus the effect of changing the diffusivity should be to vary the response of the system linearly in time and as the square of radius; a change by a factor of two implies a factor of two difference in the characteristic time, and root two in characteristic size. To put it another way: compared to an asteroid made of the higher-diffusivity material as modeled, a more realistic lower-diffusivity body would maintain its internal temperature for twice as long; or alternately, a 70 km diameter low diffusivity asteroid would be expected to have a thermal profile similar to a 100 km diameter body with twice the thermal diffusivity. The melting of an icy moon is more complicated to model, since ice close to the melting point may also transport heat by convection; but the Rayleigh number that describes the heat transport characteristics of this convection is itself inversely proportional to the thermal diffusivity. Thus it is clear that even a factor of two change in thermal diffusivity can have a profound effect on the expected thermal evolution of a small body.

Consider, as an example, the models of Kuiper Belt objects by Prialnik and Merk (2008). They assumed a body half dust and half ice by mass, with the conductivity of the ice dominated by amorphous ice between 0.3 and 0.6 W m^{-1} K^{-1}, and the conductivity of the dust represented by serpentine, set at 2 W m^{-1} K^{-1}. Bulk thermal conductivity is volume averaged; using their assumed densities for ice (0.917×10^3 kg m^{-3}) and rock (3.25×10^3 kg m^{-3}), their modeled body is 78% by volume ice. Nonetheless, the thermal conductivity of the bulk is dominated by the much more conductive rocky component. Changing the serpentine conductivity they used to our value for the CM meteorite Cold Bokkeveld, 0.28 W m^{-1} K^{-1} at their assumed starting temperature of 70 K, the bulk thermal conductivity of their body drops to half their assumed value.

Prialnik and Merk (2008) also modeled Enceladus as an object starting at 90 K made of 75% dusty material. Given these values, the resulting bulk thermal conductivity using the CM meteorite value is one-third that which they assumed. Recognizing the uncertainties in the thermal conductivity of the dusty component, Prialnik and Merk ran models assuming that their dust value was off by a factor of two; but as we have seen, the actual value may be a full order of magnitude lower than they assumed. After accounting for the dilution of this material in the ice, using our CM meteorite thermal conductivity leads to the bulk conductivity of Enceladus being 50% lower than even their "lower conductivity" case.

The evolution of Kuiper Belt objects and icy moons such as Enceladus is complex, as the models of Schubert et al. (2007) and Pri-

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alnik and Merk (2008) show, so it is not trivial to simply extrapolate what the effect of this lower thermal conductivity will be. But in general one would expect that internal melting will occur closer to the surface of these bodies, and elevated temperatures remain for a longer time, than has been calculated up to now.

The effect of low thermal conductivity on the thermal inertia (and subsequent Yarkovsky-style perturbations of asteroid orbits and spins) will be less dramatic. For one thing, as noted above, the inertia only varies as the square root of the thermal conductivity, and so even an order of magnitude change would have only roughly a factor of three change in the thermal inertia. Furthermore, the thermal inertia of solid material is almost certainly not the determining factor in the overall thermal inertia of asteroids. It is possible to derive the thermal inertia of asteroids from infrared observations (cf. Delbo et al., 2007) from which it is clear that the thermal inertia of near-Earth objects as small as a few hundred meters in diameter is dominated by dust on the surface of these objects, not on the intrinsic nature of that material itself.

However, Delbo et al. (2007) find that the effect of this dust depends on the size of the body. Extrapolating to smaller sizes, their work suggests that the thermal inertia of objects a few tens of meters across will indeed reach the levels calculated for our meteoritic material. Thus the evolution of the orbits of meteoroid fragments, from which our meteorites are ultimately derived, may depend almost entirely on the material of which the meteoroid is made. Furthermore, since there is a significant difference in the efficacy of this effect depending on the material, this may be a mechanism for controlling the rate of such evolution as a function of composition, and perhaps serve as one of the factors in the differences seen in typical cosmic ray exposure ages among the different meteorite types.

Future work will extend these measurements in two directions. First, it is important that a number of meteorites within each type are measured, to be certain that these results are truly typical of their meteorite type and not affected by unusual compositions, weathering, etc. Observing the ranges possible within a meteorite type may also provide clues as to the internal structure of the meteorite fabric, an interesting measurement in and of itself, possibly giving clues to the lithification history of the meteorites. It will be interesting to see if the rough correlation between meteorite porosity and thermal conductivity suggested by Yomogida and Matsui (1983) can be seen with more precise data. Furthermore, data taken on freshly fallen meteorites would be especially useful, as this would eliminate any effects of terrestrial weathering which tends to fill pore spaces with ferric-rich weathering products (Bland et al., 1998; Consolmagno et al., 2008). And thermal property data taken on basaltic achondrites would be invaluable for modelers of Mars, the Moon, Asteroid Vesta, and the iron meteorite parent bodies.

In addition, we are beginning a survey of meteorite heat capacities as a function of temperature. Again, knowing the range of values as a function of meteorite type and history, and how this varies with temperature, may not only provide essential data for understanding the thermal and orbital evolution of asteroidal and outer Solar System materials, but it may also be a useful way of characterizing the meteoritic material itself.

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